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Environmental Design Approach

The project incorporates strategies that support both environmental responsibility and a stronger relationship to site conditions. First, the use of locally sourced stone, such as the dark bedrock characteristic of Manhattan, grounds the architecture in the geological identity of the region while reducing the environmental impact associated with transporting materials long distances. Using materials that already exist within the local context also avoids the energy intensive processes required to synthetically reproduce similar materials.

In addition, the integration of solar panels allows the building to generate renewable energy directly on site. In a dense urban environment like Manhattan, rooftop and upper surface solar exposure can be harnesses through photovoltaic systems. For data centers, which require continuous electrical power, on site solar generation can help offset a portion of operational energy demand and reduce reliance on fossil fuel based energy sources.

Paired with solar energy generation, the project also incorporates rainwater collection as part of a broader strategy for sustainable data infrastructure. Data centers require significant cooling due to heat produced by servers. Captured rainwater can be reused for non potable uses such as toilet flushing and can also support cooling systems like evaporative towers. Together, renewable energy production and water reuse help reduce the environmental impact of data center operations by addressing both energy consumption and cooling water demand.

These strategies demonstrate how environmental systems can be integrated directly into the architectural design rather than treated as secondary infrastructure.

